Project Plan - Fintech360 Data Pipeline

1. Project Overview

This project establishes a scalable, production-grade Medallion architecture on Databricks to process loan lifecycle data from an AWS RDS source (simulated via Python Faker). The focus is on stringent data quality gates, decoupled compute/storage, and optimized serving for BI and ML workflows.

2. Timeline & Milestones (12-Week Implementation)

Phase 1: Infrastructure & Data Generation (Weeks 1)

- **Objective:** Set up foundational AWS and Databricks infrastructure and simulate transactional data.
- **Tasks:**
    - Write Python `faker` scripts to generate realistic loan applications, KYC, repayments, and DPD tracking.
    - Provision AWS RDS (PostgreSQL) and load the generated mock data.
    - Configure AWS S3 landing zone buckets and IAM policies.
    - Provision Databricks workspace and configure cluster profiles.

Phase 2: Ingestion & Bronze Layer (Weeks 1)

- **Objective:** Establish the raw data pipeline.
- **Tasks:**
    - Set up extraction from RDS to S3 Landing (CDC or micro-batch).
    - Configure Databricks Auto Loader to read incoming S3 files.
    - Build the Bronze Layer (Raw/CDC) to retain full immutable history.

Phase 3: Schema Validation & Silver Layer (Weeks 2)

- **Objective:** Cleanse, standardize, and enforce structural integrity.
- **Tasks:**
    - Implement the Schema Validation gate (verify columns, data types, nullability, and schema evolution).
    - Build the Silver Layer (Clean Data).
    - Implement PII masking and standardization of date/currency fields.

Phase 4: Data Quality & Quarantine Implementation (Weeks 3)

- **Objective:** Ensure no corrupted data enters the downstream layers.
- **Tasks:**
    - Define explicit expectations for Data Quality (Null checks, Duplicate checks, Range checks, Referential integrity).
    - Implement the routing logic (Pass/Fail).
    - Build the Quarantine (Error Monitoring) table to catch rejected records.
    - Setup alert mechanisms for high quarantine volumes.

Phase 5: Gold Layer & Governance (Weeks 3)

- **Objective:** Create business-ready datasets and secure them.
- **Tasks:**
    - Perform Business Transformation logic on Silver data (e.g., calculating collection efficiency, risk deciles).
    - Materialize the Gold Layer tables (optimized for read).
    - Configure Databricks Unity Catalog for Row-Level Security (RLS) and Column-Level Security (CLS).

Phase 6: Downstream Consumption (Weeks 4)

- **Objective:** Expose the data to BI and ML stakeholders.
- **Tasks:**
    - Connect Power BI to the Databricks SQL Warehouse/Gold tables.
    - Develop core management dashboards using query folding for performance.
    - Expose feature views to the ML models (for credit risk decisions).
    - End-to-end performance tuning and handover.

3. Resource Requirements

- **Data Engineering:** PySpark, AWS S3/RDS integration, Delta Lake management.
- **Analytics/BI:** Power BI, DAX, Databricks SQL.
- **Tools/Licensing:** AWS (RDS, S3), Databricks, Power BI Premium, GitHub (CI/CD).